RescueRoute Project Report

RescueRoute connects food donor with recipient organizations and volunteers. It takes the data from css files about donations, recipients, and volunteers, checks if the informations are correct, and creates a scheduling plan for matching them together.

Scheduling

FIFO handles the donations based on their ready time. Greedy processes donations with the minimum shelf life first to help urgent food donations before they gets expired. Each assignment check the type of food, recipient capacity, volunteer capacity, availability, number of pickups, and closing time. This help making sure the donation can be delivered to a place that accepts it.

Testing and Results

The project has passed 16 automated tests focused on input validation, matching rules, capacity limits, unavailable volunteers, FIFO order, Greedy order, integration, and metrics. The supplied data processed 4 donations: 3 was assigned, 1 was unmatched, and 130 items were assigned total. The assignment rate was 75%.

Team Work

The Data and Matching component provides the CSV data, input validation, and compatibility checks. The Algorithm component handle FIFO and Greedy scheduling. The Product and Quality component integrates the program, test it, calculates metrics, and prepares the documentation.

Run Commands

python3 main.py
python3 -m unittest discover -s tests -v